BAHIRDAR CITY ADMINSTRATION EDUCATION DEPARTMENT

CHEMISTRY MODEL EXAMINATION FOR GRADE 12 (2018 E.C)

TIME ALLOWED = 2:30 HOURS

GENERAL DIRECTIONS

This booklet contains **chemistry** model exam for grade 12. In this booklet, there are a total of **80** multiple choice questions. There is **only one best answer** for each question. Choose the best answer from the given alternatives and write the letter of your choice on the answer sheet provided. You are allowed to work on the exam questions for **2:30 hours only**.

When time is called, you must immediately stop working on the questions and put down your pen and wait for what you might be told to do. Please make sure that you have written all the required information on the answer sheet before you start to work on the exam.

GIVENS and CONSTANTS

Zn^{2+}/Zn, $E^0red = -0.76V$ $R = 0.0821 L.atm/mol.k$ $R_H = 2.18X10^{-18}J$

Cu^{2+}/Cu, $E^0red = +0.34V$ $F = 96500\frac{C}{mol} = 96500$ J/V mol

$C = 3X10^8 m/s$ $h = 6.63X10^{-34} J.s$ $n_{Al} = 3$ $K_W = 1X10^{-14}M^2$

Element	H	C	O	S	Al
Atomic Mass	1	12	16	32	27

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO!!!

2018 E.C

BAHIRDAR, ETHIOPIA

GOOD LUCK!!!

1. Which one of the following is a non-polar molecule?

   A. H_2 B. N_2 C. O_2 D. All

2. Which one of the following can **NOT** conduct electricity?

   A. NaCl(s) B. $CaCl_2$(aq) C. KCl(l) D. Band C

3. What is the relative atomic mass of an element whose isotopic composition is 90% of 30_X and 10% of 32_X? A. 32 B. 30.2 C. 30 D. 31

4. The density of a gas at a pressure of 3atm and a temperature of 300k is 10g/L. What is the molar mass of this gas? A. 90g/mol B. 71g/mol C. 82.1g/mol D. 30g/mol

5. When 20g of SO_2 react with O_2, 23g of SO_3 is formed. What is the percentage yield of the reaction?

   $2SO_2(g) + O_2(g) \longrightarrow 2SO_3(g)$

   A. 80% B. 62% C. 87% D. 92%

6. Calculate the energy emitted and the wave length of the emitted energy when an electron falls from n = 3 to n = 2 energy level.

   A. $3.63X10^{-19}$J, 547nm C. $6.211\ X10^{-19}$J, 320nm

   B. $4.321\ X10^{-19}$J, 460nm D. $3.03\ X10^{-19}$J, 656nm

7. What is the allowed values of ml for n = 4 and l = 3?

   A. -3, -2, -1, 0, 1, 2, 3 B. 0, 1, 2, 3 C. -2, -1, 0, 1, 2 D. -1, 0, 1

8. Which principle states that "**NO** two electrons in an atom can have the same four quantum numbers."?

   A. Hund's Rule B. Pauli's exclusion C. Aufbau D. Building up

9. You are given that:- I. CH_4, CCl_4 III. N_2, H_2

   II. H_2O, NH_3 IV. CH_3Cl, HCl of these molecules which one consists of polar bond and non-polar molecule? A. I, II B. II, III C. I D. IV

10. Which type of overlapping is present between C and H in CH_4?

    A. Sp – s B. sp2 – s C. sp3 – p D. sp3 – s

11. The molecule having similar electron pair arrangement with its molecular geometry is_____

    A. CH_4 B. NH_3 C. SO_2 D. H_2O

12. Which one of the following is **TRUE** about the formation of pi – bond?

    A. It is formed by overlapping of unhybridized p – orbitals above and below the inter nuclear axis.

    B. It is formed by overlapping of unhybridized p – orbitals between the inter nuclear axis.

    C. It is formed by head to head or end to end overlapping.

    D. It is formed when any two s – orbitals overlapped.

GOOD LUCK!!!

13. Which one of the following is **NOT** true according to molecular orbital theory?
    A. The number of molecular orbitals formed is equal to the number of atomic orbitals.
    B. Bonding and anti-bonding molecular orbitals have the same energy.
    C. If the bond order is equal to zero, then the molecule does **NOT** exist.
    D. A single electron in the outer molecular orbital shows that the molecule is paramagnetic.
14. A sample of gas at 27^0_c and 1atm has a volume of 3L. What volume will this gas occupy at 37^0_c and 1atm? A. 2.9L B. 3.1L C. 0.34L D. 4.1L
15. Which one of the following is the application of equilibrium constant?
    A. Predict the composition of equilibrium.
    B. Predict the direction of the reaction.
    C. Calculate the concentration of reactant and product. D. ALL
16. What will be the unit of K_p for the following reaction?

$$N_2(g) + 3H_2(g) \longrightarrow 2NH_3(g)$$

    A. Atm B. Atm^{-1} C. Atm^2 D. Atm^{-2}
17. Which one of the following is synthetic polymer?
    A. Glucose B. Cotton C. Wool D. Polyster
18. During the condensation polymerization, which molecule will release as a by-product?
    A. H_2O B. CH_3OH D. HCl D. ALL
19. Which one of the following is a thermosetting polymer?
    A. PVC B. Teflon B. Polyethylene D. Bakelite
20. What is the oxidation number of phosphorus in $Ca(H_2PO_4)_2$?
    A. -5 B. +3 C. -3 D. +5
21. Which one of the following statement is **TRUE**?
    A. Passage of current through electrolyte is accompanied by chemical change.
    B. Conductivity of electrolyte decrease with increasing temperature.
    C. Conductivity of metals increase with increasing temperature.
    D. Conductivity of metals is accompanied by chemical change.
22. Which one of the following is preferentially discharged at the anode during the electrolysis of Brine?
    A. Cl^- B. OH^- C. Na^+ D. H^+
23. What is the mass of aluminum deposited electrolytically in 193 seconds by a current of 500A?
    A. 0.11g B. 9g C. 27g D.

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Preparatory Secondary

24. Which law states that "the masses of different substances produced at each electrode by the same quantity of electricity are directly proportional their equivalent masses."?

A. Faraday's first law of electrolysis
C. Reactivity series of metal
B. Faraday's second law of electrolysis
D. Preferential discharge

25. Which one of the following is **NOT** the application of electrolysis?

A. Production of metals.
C. Refining of non-metals
B. Production of non-metals.
D. Electroplating of metals.

26. During the Haber process, which condition favors the production of NH_3?

$N_2(g) + 3H_2(g) \longrightarrow 2NH_3(g)$, ΔH = negative

A. High temperature, high pressure
C. Low temperature, high pressure
B. High temperature, low pressure
D. Low temperature, low pressure

27. Which one of the following chemical is used as a bleaching agent in sugar industry?

A. H_2SO_4 B. KOH C. NaOH D. SO_2

28. The catalyst which is used in the contact process is_____

A. Pt B. Ni C. Fe D. V_2O_5

29. The by-product in solvay process is_____

A. NaOH B. Na_2CO_3 C. NH_3 D. $CaCl_2$

30. Which law states that "matter is neither created nor destroyed in a chemical reaction."?

A. Conservation of mass
C. Multiple proportions
B. Definite proportions
D. Constant compositions

31. Which one of the following is **NOT** a use of periodic classification of elements?

A. To predict formulas of compounds.
B. To predict how to separate mixtures and substances.
C. To identify the physical properties of elements.
D. To identify the chemical properties of elements.

32. Which one of the following is **TRUE** according to kinetic molecular theory and properties of matter?

A. Gases and solids can flow.
C. Gases exert pressure in all directions.
B. Solids are extremely compressible.
D. Gases and liquids are resistant to diffuse.

33. Equal masses (20g) of CH_4 and O_2 are reacted as follows to give CO_2 and H_2O. Which one of the following is the limiting reactant? $CH_4 + 2O_2 \longrightarrow CO_2 + 2H_2O$

A. CH_4 B. CO_2 C. H_2O D. O_2

34. Which one of the following method is **NOT** used to prepare salts?

A. Reaction of active metals with acids.
B. Reaction of acids with bases.
C. Reaction of basic oxide with acids.
D. Reaction of active metals with water.

35. For the following reaction:- $3A + B \longrightarrow 2C + D$, What will be the heat of reaction based the given information $H_A = 5KJ$, $H_B = 10KJ$, $H_C = 20KJ$ and $H_D = 10KJ$?

A. 25KJ B. 50KJ C. -40KJ D. 15KJ

36. Which of the following sets of quantum numbers in an atom are acceptable?

A. 1, 0, ½ , ½ B. 3, 2, 1, 1 C. 3, 0, 0, ½ D. 2, 2, 1, ½

37. How does catalyst increase the rate of reaction?

A. By changing chemical equilibrium.
B. By increasing activation energy.
C. By lowering activation energy.
D. Can **NOT** determine.

38. Chemical kinetics are mainly concerned:-

A. Energy change that takes place in a reaction.
B. Initial and final state of chemical reaction.
C. The rate and mechanism of reaction.
D. All except A

39. For the given reaction:- $HCO_3^- + H_2O \longrightarrow H_3O^+ + CO_3^{-2}$, Which of the following are Bronsted-Lowery acids?

A. H_2O and H_3O^+ B. HCO_3^- and CO_3^{-2} C. H_2O and CO_3^{-2} D. HCO_3^- and H_3O^+

40. A solution is formed by dissolving an antiacid tablet which has a P^H of 4 at 25^0_C. What is the concentration of OH^-? A. $1X10^{-10}$ B. $1X10^{10}$ C. $1X10^4$ D. $1X10^{-4}$

41. A solution is acidic originally, and has P^H = A. Then what will be its P^H if we add more acid on it?

A. Greater than A B. Less than A C. Equals to A D. Can **NOT** determine

42. A solution contains 1M HA and 1M A^-. Then what will be the P^H of a solution when its Ka = $1X10^{-4}$?

A. 10 B. 4 C. 3 D. 11

43. Why hydrolysis of salts of strong acid and strong base gives neutral solution?

A. The anion of acid is able to react appreciably with water.
B. The cation of a base is able to react appreciably with water.
C. The anion of acid and the cation of a base are unable to react appreciably with water.
D. The anion of acid and the cation of a base are able to react appreciably with water.

44. An indicator has a **green** color in acidic form and **red** color in basic form. What will be the color of a solution containing this indicator when its P^H is 4 and KIn = $1X10^{-4}$?

A. Red B. Green C. Reddish-Green D. Black

45. A buffer solution has a $P^H = 4.5$, then what will generally be its P^H if strong acid is added to it?

A. Increase B. Decrease C. **NOT** determine easily D. Remains the same

46. What will be the conjugate base of $H_2PO_4^-$?

A. H_3PO_4 B. PO_4^{3-} C. $H_2PO_4^-$ D. HPO_4^{-2}

47. Which one of the following is **NOT** a Lewis acid?

A. CH_4 B. $BeCl_2$ C. Fe^{2+} D. CO_2

48. How many sigma and pi bonds are present in C_2H_2 respectively?

A. 3, 2 B. 3, 3 C. 2, 3 D. 4, 1

49. Which one of the following is **TRUE** about oxygen containing hydrocarbons?

A. Alcohols have high melting and boiling point than comparable molecular mass of carboxylic acids.

B. Carboxylic acids are soluble in organic solvents like benzene, ether, and alcohol.

C. Carboxylic acids are insoluble in water.

D. Carboxylic acids have equal melting and boiling point with comparable molecular mass of alcohols.

50. Which one of the following are used as sources of esters?

A. Milk, egg, fish C. Pea, chickpea, beans

B. Banana, mango, apple, honey D. Sugar, salt, water

51. Which one of the following is a 3^0 alcohol?

A. 2-Buthanol B. 1-Hexanol C. 3-Methyl-3-Hexanol D. 3-Methyl-2-Hexanol

52. Liquid oil is assumed to be best for feeding than solid (coagulated) oil. This is because liquid oil ;-

A. Has single bonds between carbon atoms and easily reacted and converted to usable energy.

B. Has multiple bonds between carbon atoms and difficult to react and converted to usable energy.

C. Has multiple bonds between carbon atoms and easily reacted and converted to usable energy.

D. Has no known scientific reason.

53. Which one of the following is **NOT** a conjugate acid–base pair?

A. NH_3 and NH_4^+ B. H_2O and OH^- C. HNO_2 and HNO_3 D. HCl and Cl^-

54. Which of the following is **TRUE** about the titration of strong acid with strong base at the equivalence point? A. $P^H > 7$ B. $P^H < 7$ C. $P^H = 7$ D. $P^H = P^{Ka}$

55. Which pairs of the following species shows a buffer solution?

A. KNO_3/HNO_3 B. NH_4Cl/HCl C. H_2O/H_3O^+ D. HCOOK/HCOOH

56. You are given that:- I. NH_3, NH_2^- II. H_2O, NH_3 III. HCO_3^-, $H2PO_4^-$ IV. H_2O, HCO_3^-, whi

pairs of species consists of both auto ionized and amphiprotic species?

A. I. II B. I, II, III C. II, IV D. II

57. NaOH is an Arrhenius base but **NOT** Bronsted-Lowery base, this is because NaOH is:-

A. Unable to donate proton. C. Able to accept proton.

B. Able to donate proton. D. Unable to accept proton.

GOOD LUCK!!!

58. Vinegar (Achetto) is used to kill microorganisms in food. What is the chemical formula of Vinegar?

A. HCOOH B. $CH_3(CH_2)_3COOH$ C. HF D. CH_3COOH

59. What will be the atom economy (in %) of the following reaction?

$CH_2 = CH_2 + H_2O \longrightarrow CH_3CH_2OH$

A. 90 B. 50 C. 75 D. 100

60. The largest reservoir of nitrogen in our planet is_____

A. Ocean B. Fossil fuel C. Biosphere D. Atmosphere

61. Which of the following pollution is an example of impacts of developmental activities on the hydrosphere? A. Air B. Water C. Land D. Soil

62. All of the following are the way to reduce air pollution except____

A. Use public transport C. Use plastic bags

B. Recycle and use D. Implement afforestation

63. Which one of the following is the cell diagram for a Daniell cell?

A. $Zn^{2+}(aq)/Zn(s)//Cu(s)/Cu^{2+}(aq)$ C. $Cu(s)/Cu^{2+}(aq)//Zn(s)/Zn^{2+}(aq)$

B. $Zn(s)/Zn^{2+}(aq)//Cu(s)/Cu^{2+}(aq)$ D. $Zn(s)/Zn^{2+}(aq)//Cu^{2+}(aq)/Cu(s)$

64. Given the reaction:- $2Au(s) + 3Ca^{2+}(aq) \longrightarrow 2Au^{3+}(aq) + 3Ca(s)$, What will be the standard free energy of a reaction at 25^0_C and $E^0_{cell} = -4.37V$?

A. 843.41KJ B. 1265KJ C. 2530KJ D. 421KJ

65. What will be the standard cell potential for $Zn(s)/Zn^{2+}(aq)//Cu^{2+}(aq)/Cu(s)$?

A. 1.1V B. -1.1V C. 0.42V D. -0.42V

66. A battery is ''dead'' if:-

A. $E_{cell} = 0$ B. $\Delta G = 0$ C. The reaction is at equilibrium. D. ALL

67. Which of the following is **TRUE** about corrosion of iron?

A. Iron rusts more quickly at low P^H C. Iron rusts more quickly in ionic solution

B. Iron does **NOT** rust in dry air. D. ALL

68. What is the importance of salt bridge in a galvanic cell?

A. Maintain electrical neutrality. C. Allow electrical conductivity.

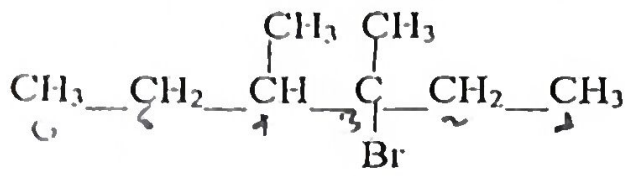
B. Prevent mixing of the two electrolytes. D. ALL

69. Which one of the following is the physical property of metals?

A. Possess positive valance. C. Are sonorous.

B. Oxidized by losing electrons. D. Usually replace by hydrogen.

70. What is the IUPAC name of the following compound?



A. 3,4-Dimethyl-4-Bromo hexane
B. 3-Bromo-3, 4-Dimethyl hexane
C. 3-Bromo-3-Methyl hexane
D. NONE

71. Which of the following compound exhibit cis-trans isomer?

A. 1-Propene B. 2-Butane C. 3-Hexene D. 3-Hexyne

72. What will be the mass (in gram) of $Al_2(SO_4)_3$ in 500ml of a solution if its concentration is 2N?

A. 57 B. 171 C. 85.5 D. 342

73. _____is the energy released when molecules of solute is generally surrounded by solvent molecules.

A. Hydration energy B. Solvation energy C. Lattice energy D. Chemical energy

74. Which law states that "at the same temperature and pressure, equal volumes of gases contain equal numbers of moles."?

A. Boyle's law B. Charles' law C. Avogadro's law D. Gay Lussac's Law

75. Which one of the following is **TRUE** about a concentration cell? A concentration cell has:-

A. Different electrodes.
B. Same electrode but different concentrations.
C. Same electrode with same concentrations.
D. **NO** current flow.

76. Which one of the following is the importance of Nernst equation?

A. To determine standard cell potential.
B. To determine cell potential at non-standard state.
C. To determine the mass of a substance.
D. To determine the current flow.

77. Which one of the following is the key difference between galvanic and electrolytic cell?

A. Both use redox reaction.
B. Both require power supply.
C. Energy source and spontaneity.
D. Both have identical electrode charges.

78. A neutral molecule having the general formula BX_3 has two lone pair of electrons on B. What will be the hybridization of B? A. sp^3 B. sp C. sp^2 D. sp^3d

79. Which one of the following alcohols yields ketones on oxidation?

A. 1^0 Alcohol B. 2^0 Alcohol C. 3^0 Alcohol D. Dihydric alcohol

80. Which of the following will **NOT** cause any atmospheric pollution?

A. Oxides of Nitrogen B. SO_2 C. O_2 D. CO



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MODEL-EXAM ONE Answer Sheet **Subject:** Chemistry

Allotted time: 1:40

Name Hanna Ayalew **Sec.** 12TMC **No.** 24

74

Direction: Blacken the circles given next to the numbers as per your answer's choice.

	A	B	C	D
1.	O	O	O	●
2.	●	O	O	O
3.	O	●	O	O
4.	O	O	●	O
5.	O	O	O	●
6.	O	O	O	●
7.	●	O	O	O
8.	O	●	O	O
9.	O	O	●	O
10.	O	O	O	●
11.	●	O	O	O
12.	●	O	O	O
13.	O	●	O	O
14.	O	●	O	O
15.	O	O	O	●
16.	O	O	O	●
17.	O	O	O	●
18.	●	O	O	Ø
19.	O	O	O	●
20.	O	O	O	●
21.	●	O	O	O
22.	●	O	O	O
23.	O	●	O	O
24.	O	●	O	O
25.	O	O	●	O
26.	O	O	●	O
27.	O	O	O	●
28.	O	O	O	●
29.	O	O	O	●
30.	●	O	O	O

	A	B	C	D
31.	O	●	O	O
32.	O	O	●	O
33.	O	O	O	●
34.	O	O	O	●
35.	●	O	O	O
36.	O	O	●	O
37.	O	O	●	O
38.	O	O	Ø	●
39.	O	O	O	●
40.	●	O	O	O
41.	O	●	O	O
42.	O	●	O	O
43.	O	O	●	O
44.	O	O	●	O
45.	O	●	O	Ø
46.	O	O	O	●
47.	●	O	O	O
48.	●	O	O	O
49.	O	●	O	O
50.	●	Ø	O	O
51.	O	O	●	O
52.	O	O	●	O
53.	O	O	●	O
54.	O	O	●	O
55.	O	O	O	●
56.	O	O	O	●
57.	O	O	O	●
58.	O	O	O	●
59.	O	O	O	●
60.	O	O	O	●

	A	B	C	D
61.	O	●	O	O
62.	O	O	●	O
63.	O	O	O	●
64.	O	O	●	O
65.	●	O	O	O
66.	O	O	O	●
67.	O	O	O	●
68.	O	O	O	●
69.	O	O	●	O
70.	O	●	O	O
71.	O	O	●	O
72.	●	O	O	O
73.	O	Ø	●	O
74.	O	O	●	O
75.	O	●	O	O
76.	●	Ø	O	O
77.	O	O	●	O
78.	O	O	O	●
79.	O	●	O	O
80.	O	O	●	O
81.	O	O	O	O
82.	O	O	O	O
83.	O	O	O	O
84.	O	O	O	O
85.	O	O	O	O
86.	O	O	O	O
87.	O	O	O	O
88.	O	O	O	O
89.	O	O	O	O
90.	O	O	O	O

	A	B	C	D
91.	O	O	O	O
92.	O	O	O	O
93.	O	O	O	O
94.	O	O	O	O
95.	O	O	O	O
96.	O	O	O	O
97.	O	O	O	O
98.	O	O	O	O
99.	O	O	O	O
100.	O	O	O	O
101.	O	O	O	O
102.	O	O	O	O
103.	O	O	O	O
104.	O	O	O	O
105.	O	O	O	O
106.	O	O	O	O
107.	O	O	O	O
108.	O	O	O	O
109.	O	O	O	O
110.	O	O	O	O

Blessed G/Michael Catholic Secondary School

Chemistry Model Exam for grade 12th.

May, 2018 Time allotted:- 2hr:30'

PHYSICAL CONSTANTS

Gas constant, R= 8.314J/mol.k=0.0821L.atm/mol.k

plank's constant= h=6.626 $x10^{-34}$J/s

Speed of light=3 x 10^{8}m/s

Rybderg constant, R_H= 2.18 $x10^{-18}$J

Faraday's constant=96500C/mol

ATOMIC NUMBER(Z) AND ATOMIC WEIGHT (A)

Elements	H	He	Be	B	C	N	O	F	Na	Mg	Al	Si	P	S	Cl	K	Ca
Z	1	2	4	5	6	7	8	9	11	12	13	14	15	16	17	19	20

I. Choose the best answer from the given alternatives.

1. Rutherford carried out experiments in which a beam of alpha particles was directed at a thin piece of metal foil. From these experiments he concluded that:

   A. electrons are massive particles.

   B. the positively charged parts of atoms are moving about with a velocity approaching the speed of light.

   C. the positively charged parts of atoms are extremely small and extremely heavy particles.

   D. the diameter of an electron is approximately equal to that of the nucleus.

2. Which is not a basic postulate of Dalton's atomic theory?

   A. Atoms are neither created nor destroyed in a chemical reaction.

   B. In a given compound, the relative number and kinds of atoms are constant.

   C. Atoms of all elements are alike, including their masses.

   D. Each element is composed of extremely small particles called atoms.

3. The e/m for positive rays in comparison to cathode rays is:

   A. very low B. high C. same D .none

4. In the following equation, what type of hybridization change occurs at the carbon atoms

   CH_3- CH_3 + $^7/_2$ O_2 $\longrightarrow$ $2CO_2$ +$3H_2O$

   A.SP^3 to SP^2 B. SP^3 to SP C. SP^3 to SP^3d D. SP^2 to SP

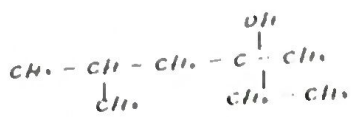
5. Element X has three isotopes :50 amu and percentage abundance of (P1),52 amu and percentage abundance of (P2),54 amu and percentage abundance of (P3)with average atomic mass =52.2,and P1=P2,find P3(the percentage abundance of the third element)

   A. 20% B. 50 % C.30% D. 40 %

6. Which element has the highest first ionization energy?

   A. Be B. B C. N D. O

7. What is the IUPAC name for the structural formula given below?



A. 2-ethyl- 4-methyl-3-pentanolp
B. 4-ethyl-2-methyl- 3- pentanone
C. 2,4- dimethyl-3-hexanol
D. 3,5-dimethyl – 3-hexanol

8. The atomic radius of an element tends to increase as you move:
A. Down a group and from right to left across a period
B. Down a group and from left to right across a period
C. Up a group and from right to left across a period
D. Up a group and from left to right across a period

9. Which of the following statement is true about oxide?
A. All basic oxide are metallic oxide
B. Amphoteric oxides are basic oxide
C. All non-metalic oxides are acidic oxide
D. Oxides are ternary compounds

10. The law of definite proportions states that:
A. Mass is neither created nor destroyed in a chemical reaction.
B. A pure compound always contains the same elements in the same proportion by mass.
C. When two elements form different compounds, the masses of one element combining with a fixed mass of the other are in a small whole-number ratio.
D. Atoms of the same element may have different masses.

11. An element has electron configuration of $[Kr]5s^2 4d^{10} 5p^2$. To which group and period respectively does the element belong in the Modern periodic table?
A. Group IVA and period 5
B. Group IIA and period 5
C. Group VA and period 7
D. Group VIIA and period 5

12. Why does the electron configuration of copper(Cu, atomic number 29) deviate from the expected $[Ar]\ 4s^2\ 3d^9$?
A. Copper prefers a full-filled 3d subshell for stability.
B. Copper has fewer protons than expected.
C. Copper's 4s orbital is filled before the 3d.
D. Copper's 4p orbital is partially filled.

13. Which of the following gives the best explanation for the fact that sodium chloride is crystalline?
A. There is strong electrostatic attraction between oppositely charged ions.
B. It contains a regular 3D arrangement of positive and negative ions.
C. The crystal contains ions.
D. There are free electrons present which reflect the light.

4. The correct order of increasing attractive strength for weak intermolecular forces is...
A. dipole-dipole interaction, hydrogen bonding, van der Waals forces
B. van der Waals forces, dipole-dipole interaction, hydrogen bonding
C. hydrogen bonding, dipole-dipole interaction, van der Waals forces
D. hydrogen bonding, van der Waals forces, dipole-dipole interaction

15. A compound is more likely to be covalent if the...
A. cation has a small size and a high charge
B. anion has a small size and a high charge
C. cation has a large size and a small charge
D. anion has a small size and a small charge

16. What are the products " A" and " B " in the reaction below?

$CH_3C \equiv CCH_3 + Cl_2 \longrightarrow A + H_2 \longrightarrow B$

A. 2,3-dichlorobutene and 2,3-dichlorobutane
C. 2,2-dichlorobutene and 2,2-dichlorobutane
B. 1,2-dichlorobutene and 1,2-dichlorobutane
D. 2-butyne and 2-butene

17. Which one of the following statements about positive ions is incorrect?

A. they are also known as cations
B. they are formed when electrons are removed from atoms
C. they are larger than the atom from which they were formed
D. they are smaller than the atom from which they were formed

18. Which of the following has the greatest covalent character?

A. LiCl
B. NaCl
C.KCl
D . RbCl

19. How many new hybrid orbitals are formed in hybridization of central atom phosphorous in PCl_5?

A.3
B. 4
C. 5
D. 2

20. Which of the following has the bonds correctly arranged in order of increasing polarity?

A. Be — F, Mg — F, N — F, O — F
C. O— F, Be — F, Mg — F, N — F
B.O— F, N — F, Be — F, Mg — F
D. N — F, Be — F, Mg — F, O — F

21. How are metallic bonds different from ionic and covalent bonds?

A. In metallic bonds, electrons are free floating. They are not transferred or shared like in ionic or covalent bonds.
B. In metallic bonds, electrons are shared. They are not free floating, like in ionic or covalent bonds.
C. In metallic bonds, electrons are transferred. They are not free floating, like in ionic or covalent bonds.
D. In metallic bonds, electrons are centralized. Electrons are decentralized in ionic and covalent bonds.

22. The rate of diffusion of two gases A and B is in the ratio of 1 : 4 and that of B and C in the ratio of 1 : 3 the rate of diffusion of C with respect to A is -

A. 1/12
B. 12
C. 6
D.4

23. According to Avogadro's law, the correct statements are:

A. Volume of gas is proportional to the no. moles at constant T and P.
B. The pressure of a gas is directly proportional to temperature of the gas under all conditions.
C. Equal volumes of different gases under similar conditions consist of equal number of molecules.
D. Equal volumes of different gases under same conditions have equal number of atoms.

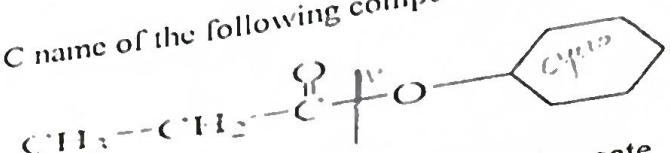
24. Which one of the following samples of gas, when sealed into a vessel of volume 0.10 m^3 is at the highest pressure?

A 1.6 g of helium (He) at 100 K
C. 1.6 g of oxygen (O_2) at 600 K
B 1.6 g of methane (CH_4) at 100 K
D. 1.6 g of sulphur dioxide (SO_2) at 1200 K.

25. The basic assumption in the kinetic theory of gases that the collisions of the gaseous molecules are perfectly elastic implies that the?

A. forces of attraction and repulsion are in equilibrium
B. gaseous molecules can occupy any available space
C. gaseous molecules will continue their motion indefinitely
D. gases can be compressed

26. The IUPAC name of the following compound will be



A. Ethylcyclohexanoate B. cyclohexylethanoate C. benzylethanoate D. ethylbenzoate

27. In the reaction:- $2H_2S + SO_2 \rightarrow 3S + 2H_2O$, the reducing agent is

A. H_2S B. SO_2 C. S D. H_2O

28. Which of the following is an example of synthesis reaction?

A. $NaCl + MgNO_3 \longrightarrow NaNO_3 + MgCl_2$

B. $K_2O + H_2O \longrightarrow 2KOH$

C. $2KClO_3 \longrightarrow 2KCl + 2O_3$

D. $2Al + 6HCl \longrightarrow 2AlCl_3 + 3H_2$

29. How many litres of NO_2 gas at STP are required to react with water to form 11.2 litres of NO gas according to the equation. $3NO_2 + H_2O \longrightarrow 2HNO_3 + NO$

A. 3.73 L B. 33.6L C. 4.31L D. 22.4L

30. In the reaction :- $N_2 + 3H_2 \rightarrow 2NH_3$, if 7 grams of N_2 gas reacts with 2 grams of H_2 gas, which one of the following is not true?

A. hydrogen is the limiting reactant

B. 8.5 grams of NH_3 is formed

C. only 0.5 gram of the excess reactant is left unreacted

D. formation of NH_3 depends on nitrogen

31. The oxidation number of phosphorous in $Na_4P_2O_7$ is

A. +6 B. +2 C. +5 D. +3

32. 3 moles of $(NH_4)_2HPO_4$ contains

A. 16 moles of nitrogen

B. 27 moles of hydrogen atoms

C. 7 moles of oxygen atoms

D. 24 moles of hydrogen atoms

33. Which of the following is not a property of solution?

A. It has a definite composition.

B. It has a homogeneous composition.

C. It is consisting of a single phase.

D. It can be physically separated into its components.

34. From CO_2, H_2O, and BeF_2, which ones have the same molecular shape

A. H_2O B. BeF_2 C. CO_2 and BeF_2 D. H_2O, BeF_2 and CO_2

35. According to molecular orbital theory, one of the following molecules does not exist. Identify it.

A. H_2 B. C_2^{+2} C. Ne_2 D. O_2^{-2}

36. For degenerate orbitals, each must be singly occupied with an electron before anyone orbital doubly occupied. This statement refers to

A. Aufbau principle B. Hund`s rule C. Pauli`s exclusion principle D. D. Heisenberg`s uncertainty princip

37. The four possible quantum numbers for the valence shell (last electron) of Potassium ($_{19}K$) is

A. $n=3, l=2, m_l=2, m_s=+1$

B. $n=4, l=0, m_l=0, m_s=+1/2$

C. $n=4, l=1, m_l=0, m_s=-1/2$

D. $n=4, l=2, m_l=1, m_s=+1/2$

38. A solution of weak base "B" has P^{H} of 9. If we add small amount of a salt containing the conjugate acid of "BH^+", which of the following statement is true about the solution?

A. The P^{H} increase and the P^{OH} decrease

B. Both the $[OH^-]$ and P^{OH} increase

C. the P^{H} decrease and the P^{OH} increase

D. $[OH^-]$ increase and P^{H} decrease

39. The conjugate acid of $C_6H_5NH_2$ is

A. $C_6H_5NH^-$ B. C_6H_5NH C. $C_6H_5NH_3^+$ D. C_6H_5NH

40. Which of the following combination **cannot** produce a buffer solution?

A. HCN/NaCN B. $HNO_3/NaNO_3$ C. $HNO_2/NaNO_2$ D. NH_3/NH_4Cl

41. Which of the following statement is **true**?

A. Addition of NaOH to a solution of HNO_3 will increase the P^{OH} of the solution

B. Addition of HCl to a solution of NaOH will decrease the P^{OH} of the solution

C. Addition of HNO_3 to a solution of H_2SO_4 will decrease the P^{OH} of the solution

D. Addition of NaOH to a solution of HCl will decrease the P^{OH} of the solution

42. Consider the following: PO_4^{3-}, HPO_4^{2-}, $H_2PO_4^-$, H_3PO_4. The term amphiprotic can be used to describe:

A. PO_4^{3-} only

B. PO_4^{3-}, HPO_4^{2-}, $H_2PO_4^-$ only

C. HPO_4^{2-} and $H_2PO_4^-$ only

D. HPO_4^{2-}, $H_2PO_4^-$, H_3PO_4 only

43. Which of the following electrolysis have the same cell reaction with the electrolysis of dilute aqueous H_2SO_4

A. Electrolysis of brine solution

B. Electrolysis of concentrated lead chloride salt

C. Electrolysis of concentrated HCl solution

D. Electrolysis of dilute NaCl solution

44. For the reaction: $Mg(s) + Ni^{+2}(aq) \rightarrow Ni(s) + Mg^{+2}(aq)$ $E^0Mg^{+2}/Mg = -2.37V$ and $E^oNi^{+2}/Ni = -0.25v$.

What is the **E^o cell** for this reaction?

A. +2.12V B. -2.62V C. -2.12v D. +2.26v

45. Using the data given below, $E^0\ Cr_2O_7^{2-}/Cr^{3+} = 1.33V$, $E^0\ Cl_2/Cl^- = 1.36V$, $E^0MnO_4^-/Mn^{2+} = 1.51V$, $E^0\ Cr^{3+}/Cr = -0.74v$, Which one of the following substances is the strongest reducing agent?

A. Cr^{3+} B. Cl^- C. Mn^{2+} D. Cr

46. When the net ionic equations: $Fe^{2+} + Cr_2O_7^{2-} \rightarrow Cr^{3+} + Fe^{3+}$ isbalanced by ion electro method in **acidic medium** , the coefficient of : Fe^{2+}, $Cr_2O_7^{(-2)}$, Cr^{3+}, Fe^{3+} become respectively

A. 6, 1, 2, 6 B. 3, 1, 2, 3 C. 5, 1, 2, 5 D. 1, 1, 2, 1

47. A solution at 25^0C contains the metal ions Ni^{+2}, Pt^{+2} and Pd^{+2} all at 1M concentrations. Consider the following standard reduction potentials $E^oNi^{+2}/Ni = -0.25V$, $E^0Pt^{+2}/Pt = 1.2V$ and $E^oPd^{+2}/Pd = 0.99V$. Which metal(s) to be plated out first when the solution is electrolyzed?

A. Ni B. Pd C. Pt D. Ni and Pd

48. A current of 1.26A is passed through an electrolytic cell containing an aqueous solution of H_2SO_4 for 7.44hrs. What volume of the gas is generated at the anode at STP? (use P= 1atm, T= 273K and R= 0.082L.atm/mol.k)

A. 2.3L B. 1.96L C. 0.76L D. 4.76L

49. For the reaction: $2A + 3B \rightarrow 4D + 5E$, the rate of the reaction in terms of $\Delta[D]$ would be written as:

A. $+4\,\Delta[D]/\Delta t$ B. $+1/4\,\Delta[D]/\Delta t$ C. $-1/4\,\Delta[D]/\Delta$ D. $+\Delta[D]/\Delta t$

50. Consider the following reaction at equilibrium. $CH_4(g) + H_2O(g) \longrightarrow CO(g) + 3H_2(g)$, $H = +260kJ$.

Which of the following is **not** the true effect on the reaction?

A. Increasing temperature favors the formation of products C. Removal of H_2 decreases the amount of CO

B. Adding CO decreases the amount of H_2 D. Decreasing pressure increases the amount of CO

51. For the reaction, $H_2(g) + Br_2(g) \longrightarrow 2HBr(g)$, $Kc = 4$ at 700 k. what is the value of kc for the reverse reactic

A. 1.4 x 10 B. 5.04 x 10 C. 1.6 x 10 D. 2.5×10^{-1}

52. For the reaction:- $C(s) + CO_2(g) \longrightarrow 2\,CO(g)$, if the partial pressure of CO_2 and CO are 2atm & 4atm respectively, what is the value of Kp at equilibrium for this reaction?

A. 8atm B. 4atm C. 0.5atm D. 32atm

53. What volume of 0.500M NaOH is required to neutralize 25.0ml of 1.2M H_2SO_4? (Assume complete ionization of the acid).

A. 60ml B. 90ml C. 100ml D. 120ml

54. Which electron transition in a hydrogen atom releases the largest energy?

A. $n=2 \rightarrow n=1$ B. $n=4 \rightarrow n=2$ C. $n=6 \rightarrow n=3$ D. $n=7 \rightarrow n=6$

55. What is the percent dissociation in 1M solution of HF containing the concentration of $H_3O^+ = 2.7 \times 10^{-2}M$?

A. 37% B. 27% C. 3.7% D. 2.7%

56. The molecular geometry of a covalent molecule with five sets of electrons from which three are bonding pair and two are lone pair is

A. Seesaw B. T - shape C. Trigonalbipyramidal D. Linear

57. Which of the following statements is TRUE?

A. Ultraviolet light has longer wavelength than visible light

B. The energy of radiation decreases as the wavelength decreases

C. The frequency of radiation increases as the wavelength decreases

D. Wave number of an electromagnetic radiation increases as wavelength increases

58. Which of the following would react to form pentylethanoate?

A. 1-propanol and pentanoic acid C. 1-pentanol and ethanoic acid

B. Ethanol and pentanoic acid D. ethanol and ethanoic acid

59. The difference between fats and oils is that:

A. Oils are liquid at room temperature. C. oils are solid at room temperature

B. Oils have more calories. D. fats are liquid at room temperature.

60. The solubility of sodium selenate, Na2SeO_4, is 84g/100g of water at 35°C. if a solution is obtained by dissolving 92 g of Na_2SeO_4 in 200g of water at 35°C, what do you call this solution?

A. diluted B. saturated C. Unsaturated D. supersaturated

61. A lab instructor is preparing 5.0 liters of a 0.10 M $Pb(NO_3)_2$ (molecular mass 331) solution. What is the mass required?

A. 165.5 g of $Pb(NO_3)_2$ and add 5.0 kg of H_2O.

B. 165.5 g of $Pb(NO_3)_2$ and add H_2O until the solution has a volume of 5.0 liters.

C. 33.1 g of $Pb(NO_3)_2$ and add H_2O until the solution has volume of 5.0 liters.

D. 33.1 g of $Pb(NO_3)_2$ and add 5.0 liters of H_2O.

62. What is the solvent in 70% alcohol solution?

A. Water B. Alcohol C. Sugar D. Kerosene

63. The indicator Bromthymol Blue (HBb) is a weak acid with $Ka = 1.0 \times 10^{-7}$ ionizes as follows:

$HBb(aq, yellow) \rightleftharpoons H^+(aq, colourless) + In^-(aq, blue)$

Which way will the equilibrium shift when NaOH is added and what will the color of the NaOH solution be containing this indicator?

A. Equilibrium will shift to the right and the color or NaOH solution will turn green

B. Equilibrium will shift to the right and the color of NaOH solution will turn blue

C. Equilibrium will shift to the left and the color of NaOH solution will turn yellow

D. Equilibrium will shift to the left and the color of NaOH solution will turn blue

64. Which is the most common ore used for the extraction of copper?

A .CuO B. $CuSO_4$ C. $CuCO_3$ D. $CuFeS_2$

65. If $NaNO_2$ is added to a solution of HNO_2, which of the following statements is true?

A. The pH of the solution will increase.

B. The pH of the solution will decrease.

C. The pH will remain the same.

D. The equilibrium will not be affected.

66. For a voltaic (or galvanic) cell using Ag, Ag^+ (1.0 M) and Zn, Zn^{2+} (1.0 M) half-cells, which of the following statements is INCORRECT? [E^O ($Ag^+/Ag = 0.8v$,) E^O ($Zn^+/Zn = -0.76$ v]

A. The zinc electrode is the anode

B. The electron will flow through the external circuit from the zinc electrode to the silver electrode.

C. The mass of the zinc electrode will decrease as the cell operates.

D. Reduction occurs at the zinc electrode as the cell operates.

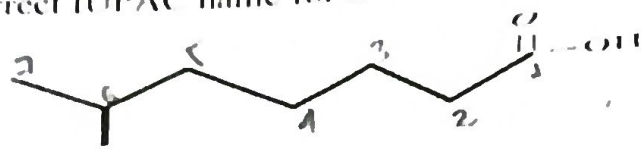
67. Bakelite is obtained from phenol by reacting with

A. HCHO B. $(CH_2OH)_2$ C. CH_3CH D. CH_3COCH_3

68. An element X has the following successive ionization energies (kJ/mol): IE_1=578, IE_2=1817, IE_3=2745, IE_4=11578. The element X is most likely:

A. Na B. Mg C. Al D. Si

69. Which of the following is the correct IUPAC name for the organic compound given below?



A. orthomethyl heptane
C. 6-methyl heptanoic acid ✓
B. 3-methyl heptanoic acid
D. 2-methyl heptanoic acid

70. Which of the following is true about salts
A. Slats are ionic compounds formed from the reaction of acidic oxide with water ×
B. Nitrate salts are water soluble salt ✓
C. The positive ions of the salt is derived from the acid ×
D. Salts conduct electricity in solid state ×

71. Phosphate and nitrate fertilizers washed-out of the soil and entered to neutral water systems, which caused water pollution, because they
A. raise the number of fishes in water ×
B. facilitate the process of depletion of oxygen which affects aquatic life ✓
C. decrease the growth of algae in water surface ×
D. suppress or blocks the eutrophication phenomena in water ×

72. A radio station broadcasts at a frequency of 100 MHz. What is the wavelength of the radio waves? (Speed of light, $c = 3.00 \times 10^8$ m/s)
A. 3.0×10^{-2} m
B. 3.0 m
C . 3.0×10^6 m
D. 3.0×10^{-3} m

73. The photoelectric effect can be observed when:
A. Light with low intensity shined on a metal surface.
B. Light of sufficient frequency shining and ejects electrons from a metal ×
C. Electrons jump to higher energy levels when exposed to light
D. The intensity of light is increased with decreasing frequency

74. The P^H of a buffer solution that contains 0.12M B and unknown concentration of BH^+ is 8. Calculate the concentration of BH^+.($K_b = 10^{-5}$)
A. 1.2M
B. 1M
C.12M
D. 0.12

75. Which of the following substances are added to natural rubber to toughen it?
A. Calcium
B. Carbon ×
C. Nitrogen
D. Sulfur

76. For lithium the enthalpy of sublimation is 161kJ /mol, and the first ionization energy is 520kJ / mol. The dissociation energy of fluorine molecule is 154kJ / mol, and the electron affinity of fluorine is 328kJ / mol. The lattice energy ofLiFis 1047kJ / mol. The overall enthalpy of formation for LiF is:
A. –617KJ/mol
B. -694kJ/mol
C. 2134KJ/mol
D. 2066kJ/mol

77. How do you prepare 250mL of 1N $Ca(OH)_2$ solution? (Ca=40, H=1, O=16)

A. By dissolving 74 g of $Ca(OH)_2$ in 500 mL of solution. C. By dissolving 37 g of $Ca(OH)_2$ in 500 mL of solution.

B. By dissolving 37 g of $Ca(OH)_2$ in 1 liter of solution. D. By dissolving 74 g of $Ca(OH)_2$ in 1 liter of solution.

78. Dental amalgam contains mercury and silver. Which one of the correct class of solution does it belong?

A. Liquid -Solid B. Liquid- Liquid C. Gas - liquid D. Solid – Solid

79. I. What volume of 2 M HCl solutions is required to react with 6g of magnesium?

$Mg(s) + 2HCl(aq) \rightleftharpoons MgCl2(aq) + H2(g)$

A. 25ml B. 50ml C. 250ml D. 400ml

80. You have 200 ml of a 0.4 M NaCl solution and you want to dilute it to exactly 0.2 M. How much water should you add?

A. 100 ml B. 200 ml C. 300 ml D. 400 ml

81. A solution was prepared by adding 48 g of methanol (CH_3OH) 81 g of water (H_2O). What is the mole fraction of methanol in this solution?

A.0.25 B. 0.75 C. 1.5 D. 4.5

82. If 2 L, 4 L and 6 L of three separate solutions of concentrations 1 M, 2 M and 3 M, respectively, are mixed together then what is the concentration of the resultant mixture?

A. 2.333 M B. 6.000 M C. 3.333 M D. 1.500 M

83. Value of Kc for a reversible reaction is 2×10^{-2} at 25°C. On increasing temperature, Kc becomes 2×10^{-3}. The reaction is:

A. Endothermic B. Exothermic C .A thermic D. Thermoneutral

84. The energies E_1 and E_2 of two radiations are 25eV and 100 eV respectively. The relationship between their wavelength (λ_1 and λ_2) will be

A. $\lambda_1 = 2\lambda_2$ B. $\lambda_1 = 4\lambda_2$ C. $\lambda_1 = \frac{1}{2}\lambda_2$ D. $\lambda_1 = \frac{1}{4}\lambda_2$

85. Green chemistry aims to?

A. Design chemical products and process that maximize profits

B. Design safer chemical products and processes that reduce or eliminate the use and generation of hazardous substance

C. Design chemical products and processes that work most efficiently

D. Utilize non-renewable energy

86. Two particles of masses m and 2m have equal kinetic energies. The de-Broglie wave lengths are in the ratio of

A. 1:1 B. 1:2 C. 1: $\sqrt{2}$ D. $\sqrt{2}$: 1

87. Which is a tertiary alcohol?

A. 2-Methylbutan-2-ol B.3-Methylbutan-2-ol C. 2-Methylbutan-1-ol D. Pentan-3-ol

88. The product of oxidation of 2-propanol with $K_2Cr_2O_7/H^+$ is:

A. Propanal B. Propanoic acid C .Acetone (propanone) D. Propene

89. A bottle of beer contains 4.5% of ethanol by volume. The density of ethanol (C_2H_5OH) is 0.79 g/cm^3).what is the molarity of ethanol?

A.0.77M B.0.077M C. 7.70M D.5.69M

90. Regarding the recycling of elements in nature which of the following statement is **correct**?

A. Nitrogen fixing bacteria convert atmospheric nitrogen gas into nitrate ion.

B. Unlike nitrogen and carbon, phosphorous is not found in atmosphere in the gaseous state.

C. Green plants convert atmospheric carbon dioxide and water to glucose by endothermic reaction

D. All of the above

BLESSED G. MICHAEL CATHOLIC SECONDARY SCHOOL

MODEL-EXAM Answer Sheet Subject: Chemistry

Allotted time: 2:10hr

77

Name Hanna Agatou Sec. 12 C No. 22

Direction: Blacken the circles given next to the numbers as per your answer's choice.

	A	B	C	D
1.	O	O	●	O
2.	O	●	O	O
3.	●	O	O	O
4.	O	●	O	O
5.	O	O	O	●
6.	O	O	●	O
7.	O	O	O	●
8.	Ø	O	O	●
9.	●	O	O	O
10.	O	●	O	O
11.	●	O	O	O
12.	●	O	O	O
13.	O	●	O	O
14.	O	●	O	O
15.	Ø	O	●	O
16.	●	O	O	O
17.	O	O	●	O
18.	Ø	O	O	●
19.	O	O	●	O
20.	O	●	O	O
21.	O	O	O	O
22.	O	●	O	O
23.	●	O	Ø	O
24.	Ø	O	O	●
25.	O	O	●	O
26.	O	●	O	O
27.	●	O	O	O
28.	O	●	O	O
29.	O	●	O	O
30.	●	O	O	O

	A	B	C	D
31.	O	O	●	O
32.	O	●	O	O
33.	●	O	O	O
34.	O	O	●	O
35.	O	O	●	O
36.	O	●	O	O
37.	O	●	O	O
38.	●	O	Ø	O
39.	O	O	●	O
40.	O	●	O	O
41.	O	O	O	●
42.	O	O	●	O
43.	O	O	O	●
44.	●	O	O	O
45.	O	O	O	●
46.	●	O	O	O
47.	O	O	●	O
48.	O	Ø	●	O
49.	O	●	O	O
50.	O	O	●	O
51.	O	O	O	●
52.	●	O	O	O
53.	O	O	O	●
54.	●	O	O	O
55.	O	O	O	●
56.	O	●	O	O
57.	O	O	●	O
58.	O	O	●	O
59.	●	O	O	O
60.	O	O	●	O

	A	B	C	D
61.	O	●	O	O
62.	O	●	O	O
63.	O	Ø	●	O
64.	O	O	O	●
65.	●	O	O	O
66.	O	O	O	●
67.	O	●	O	O
68.	O	O	●	O
69.	O	O	●	O
70.	O	●	O	O
71.	O	●	O	O
72.	O	●	O	O
73.	O	●	O	O
74.	●	O	O	O
75.	O	O	O	●
76.	Ø	●	O	O
77.	O	●	O	O
78.	●	O	O	O
79.	●	O	Ø	O
80.	O	●	O	O
81.	●	O	O	O
82.	●	O	O	O
83.	O	●	O	O
84.	O	●	O	O
85.	O	●	O	O
86.	O	O	●	Ø
87.	●	O	O	O
88.	O	O	●	O
89.	Ø	O	●	O
90.	O	O	O	●

	A	B	C	D
91.	O	O	O	O
92.	O	O	O	O
93.	O	O	O	O
94.	O	O	O	O
95.	O	O	O	O
96.	O	O	O	O
97.	O	O	O	O
98.	O	O	O	O
99.	O	O	O	O
100.	O	O	O	O
101.	O	O	O	O
102.	O	O	O	O
103.	O	O	O	O
104.	O	O	O	O
105.	O	O	O	O
106.	O	O	O	O
107.	O	O	O	O
108.	O	O	O	O
109.	O	O	O	O
110.	O	O	O	O
111.	O	O	O	O
112.	O	O	O	O
113.	O	O	O	O
114.	O	O	O	O
115.	O	O	O	O